Histograms and Polygons

Data Analysis

2026

Preliminary Mathematics Standard

Outline

Table of Contents

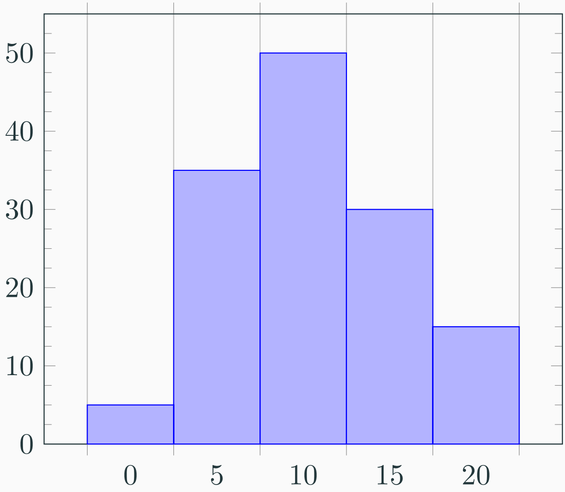
Histogram

A histogram is a type of visual summary used in statistics. It uses columns that correspond to the scores in the frequency distribution table.

When constructing a histogram:

- each column is centred on the score (these values are called the bins)
- the columns join up with no gaps

Frequency Histogram



Cumulative Frequency Histogram

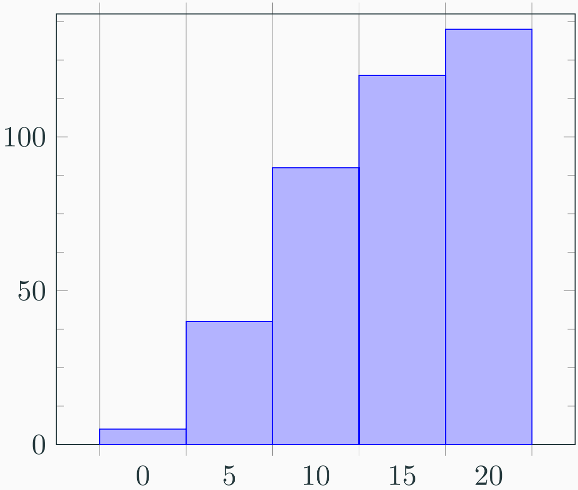


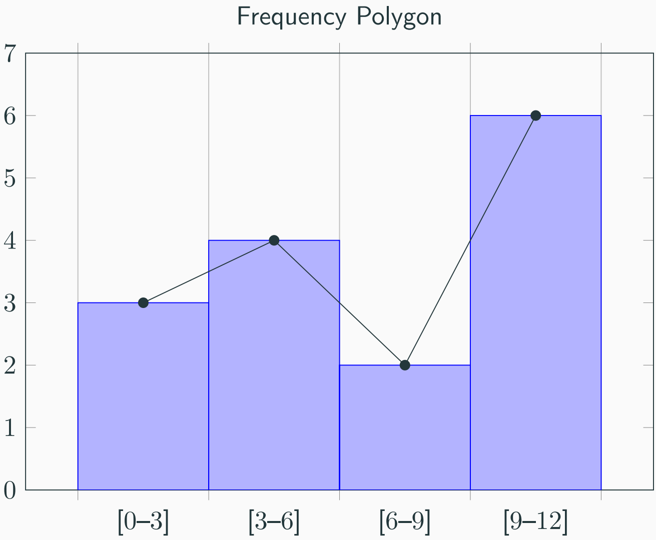
Table of Contents

Polygon

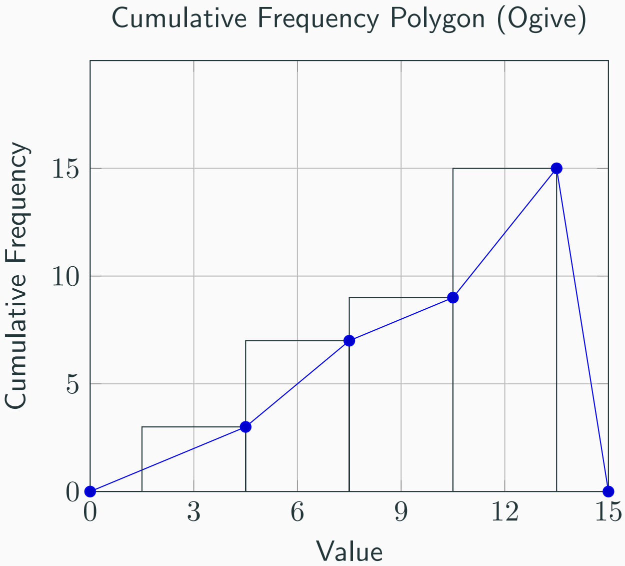
A polygon is the other common type of visual summary used in statistics. It is a line graph.
When constructing a polygon:

- the plotted points on a frequency polygon are at the centre top of each column
- start the polygon at the previous bin and finish at what would be the next bin
- A cumulative frequency polygon starts at the bottom left vertex of the first column and then continually connects to the top right vertex of each column

Frequency Polygon



Cumulative Frequency Polygon (Ogive)



Histogram with Polygon Overlay

A frequency polygon can be drawn **on top of** a histogram by connecting the midpoints of each column's top edge.

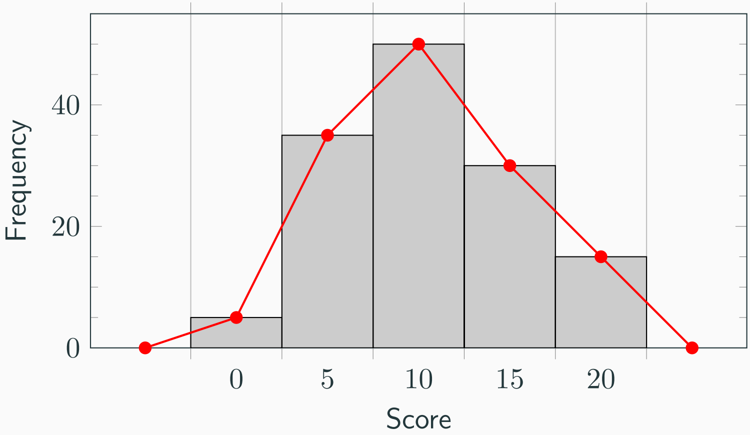


Table of Contents

Grouped Data

When data is continuous or the number of different discrete pieces of data is large, it is useful to group the data into class intervals (classes).

Guide to grouping data:

- classes must be inclusive and non-overlapping
- each score must belong to one, and only one class
- any scores on a boundary, must be treated the same for all classes

  i.e. always place in the lower class or always place in the upper class and note how the data has been handled
- determine the number of classes, generally it should be between 5 and 10
- intervals should be the same width determined by:

$$\text{width} = \text{range} \div \text{number of intervals}$$

Why Do We Group Data?

Some datasets need to be grouped to allow for appropriate representation and analysis:

- **Continuous data** can take any value in a range (e.g. heights, times) — listing every exact value is impractical
- **Large discrete datasets** with many different values become unreadable in a raw frequency table
- Grouping **reveals patterns** such as spread, centre, and shape that are hidden in raw data
- Grouped data allows us to construct meaningful **histograms and polygons**

Trade-off

Grouping loses some precision — individual scores are replaced by their class centre for calculations.

Grouped Data

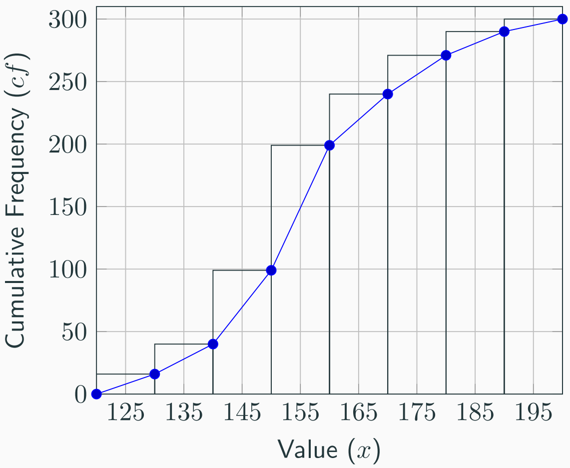
A sample of the heights, in centimetres, of 300 students has been placed into eight classes.

Height (cm)	120-129	130-139	140-149	150-159	160-169	170-179	180-189	190-199
# of people	16	24	59	100	41	31	19	10

Using a frequency distribution table, find the mean, median class, modal class, variance and standard deviation.

Example
Use the ogive to calculate the median.

Height of Students



Linear Interpolation

Table of Contents

Selecting the Best Graph for a Dataset

Different graphs suit different types of data. Choosing the right graph makes data easier to interpret.

Graph Type	Best Used For
Histogram	Continuous or grouped data; showing distribution shape
Frequency Polygon	Comparing two or more distributions on the same axes
Stem-and-Leaf Plot	Small datasets; retains original values; shows shape
Bar Chart	Categorical (non-numerical) data
Divided Bar Chart	Comparing parts of a whole across categories
Pie Chart	Showing proportions of a whole (few categories)

Justifying Your Choice of Graph

When asked to **justify** why a graph is appropriate, consider:

- **Type of data** — is it categorical, discrete, or continuous?
- **Size of dataset** — large datasets need grouping; small ones can use stem-and-leaf or dot plots
- **Purpose** — are you comparing groups, showing a distribution, or displaying proportions?

Example
The heights of 300 students are recorded. Justify the use of a histogram.

Justifying Your Choice of Graph

When asked to **justify** why a graph is appropriate, consider:

- **Type of data** — is it categorical, discrete, or continuous?
- **Size of dataset** — large datasets need grouping; small ones can use stem-and-leaf or dot plots
- **Purpose** — are you comparing groups, showing a distribution, or displaying proportions?

Example
The heights of 300 students are recorded. Justify the use of a histogram.

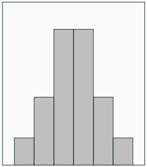
Heights are continuous data with a large number of values. A histogram allows the data to be grouped into class intervals, revealing the shape of the distribution and making analysis practical.

Table of Contents

Shape of a Distribution

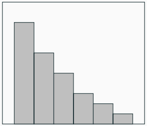
The **shape** of a distribution describes how the data is spread. We identify three main shapes:

Symmetric



Mean $\approx$ Median $\approx$ Mode

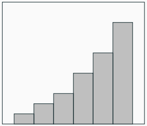
Positively Skewed



Tail extends to the **right**

Mean $>$ Median $>$ Mode

Negatively Skewed



Tail extends to the **left**

Mean $<$ Median $<$ Mode

Identifying Distribution Shape

How to identify the shape

- Look at where the **tail** (the longer, thinner end) points
- A tail pointing **right** $\Rightarrow$ positively skewed (skewed right)
- A tail pointing **left** $\Rightarrow$ negatively skewed (skewed left)
- **No tail** (roughly mirror image) $\Rightarrow$ symmetric

Example
Describe the distribution: exam scores where most students scored highly but a few scored very low.

Identifying Distribution Shape

How to identify the shape

- Look at where the **tail** (the longer, thinner end) points
- A tail pointing **right** $\Rightarrow$ positively skewed (skewed right)
- A tail pointing **left** $\Rightarrow$ negatively skewed (skewed left)
- **No tail** (roughly mirror image) $\Rightarrow$ symmetric

Example

Describe the distribution: exam scores where most students scored highly but a few scored very low.

The distribution is negatively skewed. Most scores are clustered at the high end, with a tail extending to the left towards the lower scores.

Table of Contents

Stem-and-Leaf Plots

A **stem-and-leaf plot** organises data so that individual values are retained.

- The **stem** represents the leading digit(s)
- The **leaf** represents the final digit
- Data is read left to right in order

Example
The following scores out of 50 were recorded:

21, 25, 28, 31, 33, 33, 37, 42, 44, 48

Stem-and-Leaf Plots

A **stem-and-leaf plot** organises data so that individual values are retained.

- The **stem** represents the leading digit(s)
- The **leaf** represents the final digit
- Data is read left to right in order

Example
The following scores out of 50 were recorded:

21, 25, 28, 31, 33, 33, 37, 42, 44, 48

Stem	**Leaf**
2	1 5 8
3	1 3 3 7
4	2 4 8

Key: $2 \mid 1 = 21$

Back-to-Back Stem-and-Leaf Plots

A **back-to-back stem-and-leaf plot** compares two datasets using a shared stem.

- One dataset's leaves go to the **left**, the other to the **right**
- Read the left side from the stem outward (right to left)
- Useful for comparing the shape, spread, and centre of two groups

Example
Test results for Class A (left) and Class B (right):

Class A	Stem	Class B
8 5 2	5	1 4 6
9 7 4 1	6	2 5 8
6 3	7	0 3 7 9
5	8	1 4

Divided Bar Charts

A **divided bar chart** (also called a segmented bar chart) shows how a total is split into categories.

- Each bar is divided into segments proportional to their frequency or percentage
- Useful for comparing the composition of groups
- Can display actual values or percentages

Example

A school of 200 students: 80 study Science, 70 study Arts, 50 study Commerce.

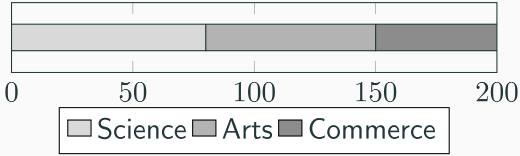


Table of Contents

Analysing a Statistical Infographic

A **statistical infographic** combines graphs, numbers, and visuals to communicate data to a broad audience. When analysing an infographic, consider:

- **What data is displayed?** — identify the dataset and its context
- **Which graph types were chosen?** — e.g. pie chart, bar chart, pictograph
- **Why were those graphs chosen?** — justify based on data type and purpose
- **What conclusions can be drawn?** — interpret trends, comparisons, and proportions

Justifying graph choice in an infographic
For example: a pie chart is appropriate when comparing proportions of a whole with few categories, as it makes the relative sizes immediately visible.

Table of Contents

Limitations of Graphical Representations

Graphs can be misleading or limited in the conclusions they support.

Common limitations:

- **Grouped data loses detail** — replacing scores with class centres introduces error into calculations
- **Graphs hide outliers** — extreme values may be absorbed into a bin without being obvious
- **Sample size not shown** — a graph may look balanced even if n is very small
- **Correlation is not causation** — a trend in a graph does not mean one variable causes another
- **Predictions beyond the data** — extrapolating beyond the range of data may be unreliable

How Graphs Can Mislead

A graphical representation can lead to **misinterpretation** when:

- The y-**axis does not start at zero** — small differences appear much larger than they are
- **Unequal bin widths** are used in a histogram without adjustment — bars appear to overrepresent some intervals
- The **scale is inconsistent** or broken — making trends look steeper or flatter
- **Pictographs use differently sized symbols** — making comparisons inaccurate
- **Data is cherry-picked** — only showing a time period or subset that supports a particular conclusion

Always ask:
Does the graph accurately represent the data, or has it been designed to create a particular impression?

Making Conclusions and Predictions

When interpreting a graph to draw conclusions:

- State what the data shows **specifically** (e.g. "the modal class is 150–159 cm")
- Note any **trends** or **patterns** (e.g. scores increasing over time)
- Acknowledge **limitations** before making predictions (e.g. small sample size, grouped data)
- Predictions **within** the data range (interpolation) are more reliable than those **beyond** it (extrapolation)

Example
Based on the ogive for student heights, we estimate the median height is approximately 157 cm. This prediction is reliable as it lies within the range of collected data, though grouping the data into 10 cm classes means our estimate may differ slightly from the true median.

Table of Contents

Homework

Complete all questions from *Data Analysis — Histograms & Polygons*.